Computer Science & DA

Probability and Statistics

Permutation and Combination

Lecture No. 03

By- Dr. Puneet Sharma Sir

Recap of previous lecture

Topic Permutation and Combination-02

$7 \times 6 \times 5 \times 4 \times 3 = {}^7P_5 = {}^7C_5 \times 5!$

Topics to be Covered

Topic Permutation and Combination-03

(HW) Sol: $\underbrace{\text{Ball } B_i \text{ is not placed in cell } C_i \ \forall i}_{\text{Dearrangements}}$ & $\underbrace{\text{No cell remains empty}}_{\text{R.H.A}}$?

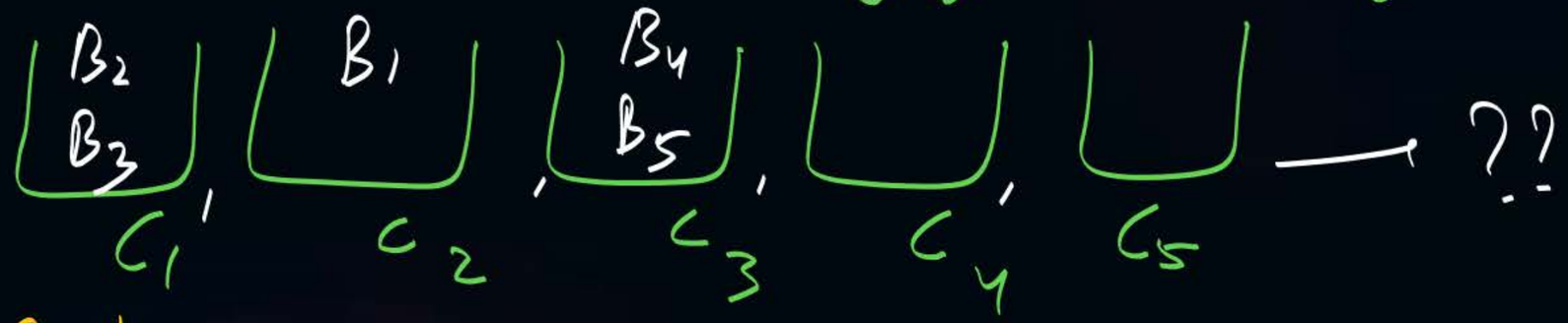
$$\text{Ans for } (n=5) = 5!\left[1-\frac{1}{1!}+\frac{1}{2!}-\frac{1}{3!}+\frac{1}{4!}-\frac{1}{5!}\right] = \cdots = 44$$

Note:- All correct or two wrong or three wrong or 4 wrong or all wrong = Total

$$\Rightarrow \overset{\shortparallel}{1} + {}^5C_2 \times (1) + {}^5C_3 \times (2) + {}^5C_4 \times (9) + {}^5C_5 \times (44) = 5!$$

$$\Rightarrow 1 + 10 + 20 + 45 + 44 = 120$$

Note → In previous Question; following types of Dearrangements are not included in Ans $= 44$



with in bracket

Some facts: →

1. Always together (= Not separated) → Assume them as one unit.
2. All Never together (= All do not come together) → Total − Always together.
3. No two Boys are seated together → First arrange Girls
4. Alternately → Two Cases arises (In linear arrangement.)
5. Particular (= fix) → No Need to arrange or Select
6. Alice → use the Result $\frac{n!}{p!\,q!\,r!}$

E.Q. 4 Boys and 4 Girls are to be seated in a Row then find the possible number of arrangements

if (1) there is No Restriction = ? = 8!
(RNA)

(2) All Boys and All Girls are together = ?

$(\underset{1}{B_1B_2B_3B_4}), (\underset{2}{G_1G_2G_3G_4}) = \underset{(P)}{2!} \times \underset{(B)}{4!} \times \underset{(G)}{4!}$
$= 1152$

(3) All G are not separated = ?

$\underset{1}{B_1}(\underset{2}{G_1G_2G_3G_4})\underset{3}{B_2}\underset{4}{B_3}\underset{5}{B_4} = \underset{(P)}{5!} \times \underset{(G)}{4!} = 2880$

(4) All G' are Never together = ? = Total − G always together
$= 8! - 5! \times 4! = 37440$

(5) No two Girls are seated together = ?

= First arrange Boys

✓ B_1 ✓ B_2 ✓ B_3 ✓ B_4 ✓ $= \underset{(B)}{4!} \times \underset{(G)}{{}^5P_4}$
$= 2880$

(*) Part 5 is a particular Case of Part (4)

(*) Various arrangements

$(\underset{1}{G}\underset{2}{B}\underset{3}{G}\underset{4}{B}\underset{5}{G}\underset{6}{B}\underset{7}{G}\underset{8}{B}) \rightarrow 4! \times 4!$

or

$(\underset{1}{B}\underset{2}{G}\underset{3}{B}\underset{4}{G}\underset{5}{B}\underset{6}{G}\underset{7}{B}\underset{8}{G}) \rightarrow 4! \times 4!$

$(G\ B\ G\ \underline{B\ B}\ G\ B\ G) \rightarrow$??

Ans = 2880

(6) Boys and Girls are to be seated alternately?

$(B_1\ G_1\ B_2\ G_2\ B_3\ G_3\ B_4\ G_4) \longrightarrow 4! \times 4!$

OR

$(G_1\ B_1\ G_2\ B_2\ G_3\ B_3\ G_4\ B_4) \longrightarrow 4! \times 4!$

Ans $= 2 \times 4! \times 4! = 1152$

(*) Part (5) is a particular case of part (5)

(7) In 4 girls there are two sisters then find the number of arrangements in which two sisters are not together = ?

Sol: Sisters are together =

$\underset{1}{B_1}\ \underset{2}{B_2}\ \underset{3}{(S_1\ S_2)}\ \underset{4}{B_3}\ \underset{5}{G_1}\ \underset{6}{B_4}\ \underset{7}{G_4}$

$= 7! \times 2!$

So Two Sisters are not together

= Total − Sisters always together

$= 8! - 7! \times 2!$

(8) there is exactly one Girl b/n two sisters = ? $= {}^2C_1 \times 1 \times 6! \times 2!$

$\underset{1}{(S_1\ G\ S_2)}\ \underset{2}{G}\ \underset{3}{B}\ \underset{4}{B}\ \underset{5}{B}\ \underset{6}{B}$

(9) there is exactly one person b/w two sis

$\underset{1}{(S_1\ P\ S_2)}\ \underset{2}{P}\ \underset{3}{P}\ \underset{4}{P}\ \underset{5}{P}\ \underset{6}{P} = {}^6C_1 \times 6! \times 2!$

(10) there is $\boxed{\text{particular Boy}}$ b/n two sisters = ?

fix

$$\underset{1}{G}\ \underset{2}{B}\ \underset{3}{(S_1\ B\ S_2)}\ \underset{4}{B}\ \underset{5}{B}\ \underset{6}{G} = {}^1C_1 \times 1 \times 6! \times 2!$$

(11) Two sisters are separated by there Bro?

$$\underset{1}{(S_1\ B\ S_2)}\ \underset{2}{B}\ \underset{3}{B}\ \underset{4}{B}\ \underset{5}{G}\ \underset{6}{G} = 6! \times 2!$$

No Need to select Brother bcz he is particular

(12) Elder and younger sister are to be seated at 1st & last position resp. $= 6!$

$$(S_1)\ \underbrace{G\ G\ B\ B\ B\ B}\ (S_2)$$

(13) Elder and younger sisters are to be seated at extreme positions ?

$$S_1\ (G\ G\ B\ B\ B\ B)\ S_2 = 6! \times 2!$$

(14) Two sisters are to be seated at Extreme Positions but together ?

Case I: $(S_1 S_2)\ G\ G\ B\ B\ B\ B = 6! \times 2!$

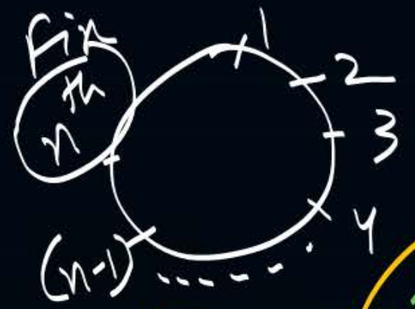
or

Case II: $G\ G\ B\ B\ B\ B\ (S_1 S_2) = 6! \times 2!$

$$Ans = 2 \times 6! \times 2!$$

(1) Number of Linear arrangements of n different things $= n!$

(2) " " Circular " " " n different things $= (n-1)!$



(3) " " Linear " of n things in which p are alike, q are alike $= \dfrac{n!}{p!\,q!}$

$= \dfrac{{}^nP_n}{p!\,q!}$

(*) Form circle only once in a given Question.

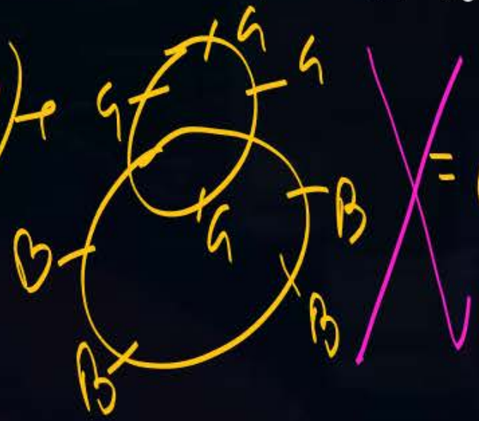
(*) In Case of circular arrangement if we have Alternately → only one Case will arise.

F.Q. 4 Boys and 4 Girls are to be seated around a circular Table for tea Party. then find Number of arrangements if (1) there is No Restriction $= ? = (8-1)!$

(2) Girls are not separated $= ?$

$= \underset{(C.A)}{(5-1)!} \times \underset{(L.A)}{4!}$

WRONG APP →



$= (5-1)! \times (4-1)!$

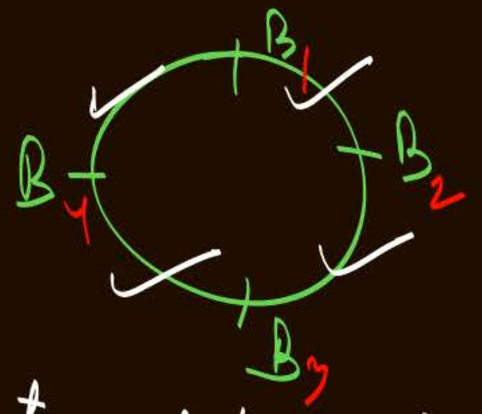
(3) All G are never together = ?

= Total − G always together

$= (8-1)! - (5-1)! \times 4!$

(4) No two 'G' are together = ?

≈ First arrange Boys in a circle



$= (4-1)! \times 4!$

Note while in L.A. :- $\checkmark B \checkmark B \checkmark B \checkmark B \checkmark = 4! \times {}^5P_4$

(5) B & G are seated alternately = ?

≈ Same as part (4)

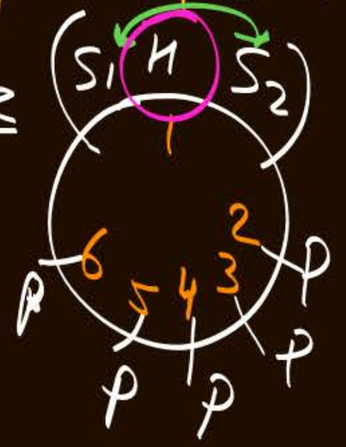
Note: while in case of L.A. ⇒ BGBGBGBG $= 4! \times 4!$

GBGBGBGB $= 4! \times 4!$

$= 2 \times 4! \times 4!$

(6) Two sisters wants to be seated at the adjacent sides of Host ?

Sol:-



$= (6-1)! \times 2!$

F.Q 3 → How many 11 letters words can be formed using the letters of the word.

'MATHEMATICS' if ① there is No Restriction = ? $= \frac{11!}{2!\,2!\,2!}$ A, A, E, I

MM AA TT E H I C S $(n = 11)$

1 2 3 4 5 6 7 8 9 10 11

② Vowels are not separated = ?

MM (AAEI) TT H C S $= \frac{8!}{2!\,2!} \times \frac{4!}{2!}$

1 2 3rd 4 5 6 7 8

③ All vowels are together & all consonants are together = ?

(AAEI), (MMTTHCS) $= 2! \times \frac{4!}{2!} \times \frac{7!}{2!\,2!}$

1 2

④ No two Vowels are together = ?

= first arrange Consonants

_M_M_T_T_H_C_S_

$= \left(\frac{{}^{7}P_7}{2!\,2!}\right) \times \left(\frac{{}^{8}P_4}{2!}\right)$

Consonants Vowels.

(5) All vowels are never together = ?

= Total − Vowels always together

$$= \frac{11!}{2!2!2!} - \left(\frac{8!}{2!\times 2!} \times \frac{4!}{2!}\right)$$

(6) Vowels occupy Even positions: →

C	O	C	O	C	O	C	O	C	O	C
O	E	O	E	O	E	O	E	O	E	O
1	2	3	4	5	6	7	8	9	10	11

$$= \frac{{}^5P_4}{2!} \times \frac{{}^7P_7}{2!\times 2!}$$

$\underbrace{AAEI}_{4V}$, $\underbrace{M,M,T,T,H,C,S}_{7C}$

TEST

PQ In how many ways 7 plus sign and 5 minus sign can be arranged in a row so that no two minus sign are together?

Sol: first arrange +ve sign.

✓ + ✓ + ✓ + ✓ + ✓ + ✓ + ✓ + ✓ $= \frac{{}^7P_7}{7!} \times \frac{{}^8P_5}{5!}$

$= {}^8C_5 = {}^8C_3 = 56$ Ans.

7(+) & 5(−)

(ii) Total arrangements (w/o any restriction) $= \dfrac{12!}{7!\,5!}$

So Req Prob $= \dfrac{f}{T} = \dfrac{56}{\left(\dfrac{12!}{7!\,5!}\right)} = ?$

(*) $2 + 3 = 5$

$2 + 3 = 5$

THANK - YOU